Where does the model error come from?

A benchmark without a network, residual diagnostics, and a plan

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In brief

1. **The network is not the limitation.** A fit with 23 free numbers per m_{hh} bin and no network, the best any model of the form $\sigma = \sum_i C_i(\theta) a_i(m_{hh})$ can do, matches the fine-tuned ensemble on the test spectra (median χ^2/N 1.088 vs 1.085; 1.58 vs 1.65 per bin below 555 GeV). A better network cannot remove the model error.
2. **The excess is in the agreement between POWHEG and the 23-term form.** It is not count statistics, not the χ^2 definition, not the normalisation, and not the statistical uncertainty of the fit.
3. **It grows with cancellation and is smooth in m_{hh}.** Spectra whose cross section is a small difference of large terms have $\langle z^2 \rangle \approx 1.9$, and their residuals are correlated between neighbouring bins (0.56); spectra without cancellation agree with the 23-term form to within Poisson noise ($\langle z^2 \rangle \approx 1.02$–$1.08$).
4. **A better error model follows.** A discrepancy proportional to the size of the individual terms, $\varepsilon(m_{hh}) \sum_i |C_i a_i|$, with $\varepsilon \approx 0.1$–0.9%, calibrates every degree of cancellation (0.97–1.06), where the current $a(m_{hh})\sigma_k^2$ under-covers the strongest-cancellation fifth (1.52).
5. **The likely origin is the generator**, a small relative inaccuracy of each coupling component, for example from the interpolated two-loop virtual amplitudes. This is a question for the people who produced the samples.

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1 A benchmark without a network

In bin k of every spectrum, `HEFTNet` predicts $\sum_{i=1}^{23} C_i(\theta)\, a_i(m_k)$: the cross section is *linear* in the 23 numbers $a_i(m_k)$. So the best possible values of those numbers, for this functional form and these data, follow from a weighted linear least-squares fit in each bin separately, with no network, no smoothness and nothing shared between bins:

$$\hat{\boldsymbol{a}}_k = \arg\min_{\boldsymbol{a}} \sum_s \frac{\left(t_{sk} - \boldsymbol{C}(\theta_s)\cdot\boldsymbol{a}\right)^2}{\sigma_{sk}^2}, \qquad \sigma_{sk}^2 = q_s(t_{sk} + q_s), \tag{1}$$

over the about 3,400 training spectra that contribute to bin k. With 23 parameters per bin and thousands of spectra, the statistical uncertainty of $\hat{\boldsymbol{a}}_k$ is small (it is included below). Any model of the form $\sum_i C_i a_i$, a network or anything else, can at best reproduce this fit; a model that smooths in m_{hh} can only do worse on the training data.

	median χ^2/N	$\langle\chi^2\rangle$, 255–555 GeV	>555 GeV	spectra $\chi^2/N>5$
free fit per bin (no network)	1.088	1.58	1.121	4
fine-tuned ensemble	1.085	1.65	1.115	4

Table 1: The 1,000 test spectra, with the exact Poisson noise $q_s t_{sk}$. The worst spectra are the same for both (test spectrum 815: $\chi^2/N = 57$ and 68).

The network already sits at the floor set by the data (table 1, fig. 1 left). On the training rows themselves the free fit gives 1.37 per bin below 555 GeV: the excess is not a failure to generalise.

2 What the floor is not

- **Not small-count statistics.** Near threshold the bins hold about 1,000–1,800 events (median), so the Gaussian approximation is excellent. The excess is the same with the data variance (Neyman), the model variance (Pearson) or the Poisson deviance: 1.58, 1.56, 1.54 below 555 GeV for the free fit.
- **Not the normalisation.** The ratio of each spectrum's total to the fit's total scatters by 0.47%, as expected from the counts alone (0.51%): POWHEG's integration error on σ_{tot} is negligible.
- **Not the statistical uncertainty of the fit.** Adding the prediction variance of the free fit, $\boldsymbol{C}^{\mathsf{T}}\,\mathrm{Cov}(\hat{\boldsymbol{a}}_k)\,\boldsymbol{C}$, lowers the mean χ^2 per bin below 555 GeV only from 1.58 to 1.46; that variance is typically 5% of the noise (12% in the most cancelling spectra).

3 What the floor is

Smooth deviations. Poisson noise, and errors of the per-bin fit, are independent from bin to bin. The residuals are not (fig. 1 right): below 555 GeV neighbouring bins are correlated by 0.27, falling to zero over about 8 bins (120 GeV); above 555 GeV a small correlation of about 0.07 persists at all separations. POWHEG departs from the 23-term form by a smooth, spectrum-dependent amount.

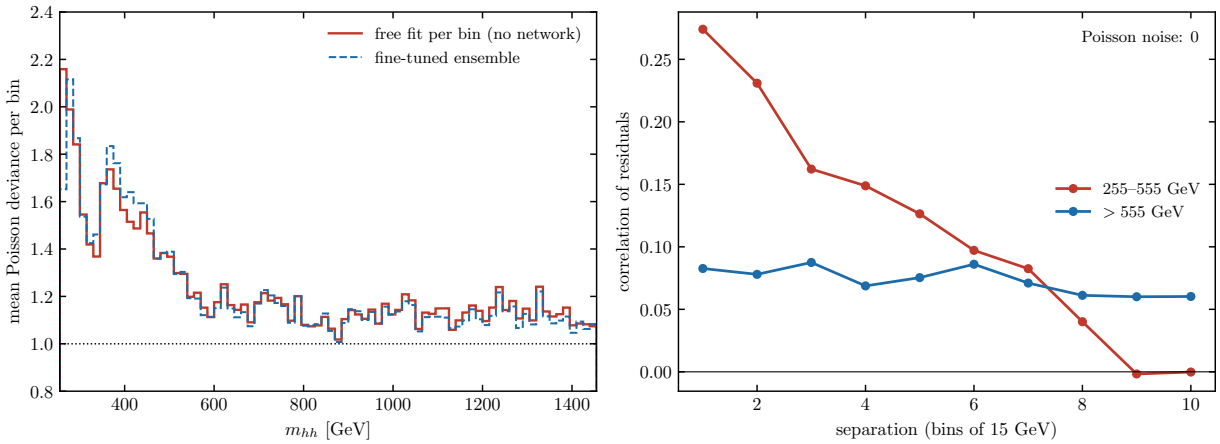


Figure 1: Left: mean Poisson deviance per bin on the test spectra, for the free fit and the ensemble; 1 is the value expected from Poisson noise alone. Right: correlation between the residuals $z = (n - \hat{n})/\sqrt{\hat{n}}$ of two bins of the same spectrum, as a function of their separation. Poisson noise gives zero.

cancellation quintile (1 = strongest)	1	2	3	4	5
$\langle z^2 \rangle$ (free fit, noise only)	1.89	1.12	1.11	1.14	1.08
lag-1 residual correlation, 255–555 GeV	0.56	0.14	0.09	0.04	0.01
lag-1 residual correlation, $>$ 555 GeV	0.24	0.05	0.05	0.02	0.01

Table 2: Excess and smoothness of the residuals by degree of cancellation (200 test spectra each).

Largest where the cross section cancels. Define the cancellation ratio of a spectrum as $\sigma_{\text{tot}} / \sum_k \sum_i |C_i a_{ik}|$: 1 when all terms add, small when the cross section is a small difference of large terms. The excess depends strongly on it (fig. 2 left, table 2):

The test spectrum with the strongest cancellation (815, ratio 0.0025: the cross section is 0.25% of the sum of the magnitudes of its terms) is also the worst fitted, by the free fit and the network alike, with a smooth shape that neither can follow (fig. 2 right). Where there is no cancellation, POWHEG agrees with the 23-term form almost to Poisson precision.

Interpretation

A small *relative* error on each of the 23 coupling components of the POWHEG cross section, smooth in m_{hh}, is invisible when the components add, but is amplified when they cancel: an error of 0.5% on terms that cancel to 0.25% of their size is a 200% error on the result. Such an error is a property of the samples, not of the network. Independent noise, for example from negative-weight events, would not produce correlated residuals and is disfavoured.

4 A better model of the error

The interpretation suggests replacing the discrepancy term $a(m_{hh})\,\sigma_k^2$ by one proportional to the size of the individual terms:

$$s^2_{\text{disc}} = \varepsilon(m_{hh})^2 \Big(\sum_{i=1}^{23} |C_i(\theta)\, a_i(m_{hh})| \Big)^2 . \tag{2}$$

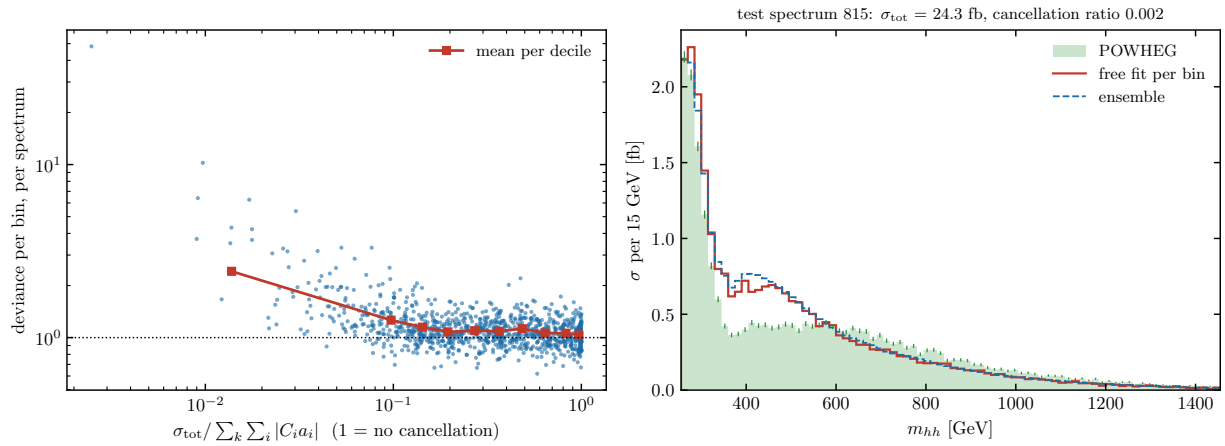


Figure 2: Left: mean deviance per bin of each test spectrum against its cancellation ratio; squares: means in deciles. Right: the worst test spectrum (815), POWHEG with the free fit and the ensemble.

Fitted bin by bin on one half of the test spectra and checked on the other half (as a quick test; for the paper it would be fitted on the quick-test rows):

		$\langle z^2 \rangle$		by cancellation quintile				
discrepancy	all	255–555	> 555	1	2	3	4	5
$a(m_{hh})\,\sigma_k^2$ (current)	1.054	1.157	1.019	1.52	0.95	0.95	0.95	0.90
$\varepsilon(m_{hh})^2(\sum_i \lvert C_i a_i \rvert)^2$	1.013	1.043	1.003	0.98	0.97	1.04	1.06	1.02

Table 3: Calibration of two discrepancy models on a held-out half of the test spectra, with the free fit as the model.

The new form calibrates every degree of cancellation; the current one is too narrow for the most cancelling spectra and slightly too wide for the rest. The fitted ε is 0.1–0.2% near threshold and up to about 0.9% at higher m_{hh}: a plausible size for the relative precision of a numerical calculation.

5 Plan

1. **Ask about the generator** (the colleague who produced the samples, and, if needed, the authors of the POWHEG `ggHH` implementation):
   - How were the two-loop virtual amplitudes evaluated (interpolation grids, which version), and what is their quoted relative precision near threshold?
   - Was each HEFT point an independent run with its own random seed and integration grids, or were spectra obtained by reweighting from a basis?
   - Were negative-weight events produced, and how were they treated?

   If a few spectra with strong cancellation could be regenerated with higher precision, a direct comparison would settle the origin.
2. **Adopt the cancellation-aware discrepancy** for the paper: fit $\varepsilon(m_{hh})$ on the quick-test rows with the Laplace mixture as the model, re-validate on the test spectra, and update the appendix, the figures and the summary.

3. **Make the discrepancy smooth in m_{hh}**: a Gaussian-process term with amplitude $\varepsilon \sum_i |C_i a_i|$ and a correlation length fitted to the residuals (fig. 1 suggests 5–8 bins near threshold), in the spirit of Kennedy and O'Hagan. This is the principled version and matters for any use of the spectra that sums bins.
4. **Keep the network.** The free fit shows nothing to gain in χ^2. Remaining improvements to the model are about how well the individual a_i are determined (the poorly constrained a_{16}–a_{20}), e.g. by fitting in a better-conditioned basis of the 23 monomials; they matter for interpreting the a_i, not for the cross section.
5. **Keep the free fit as a benchmark** (`docs/modelerror/`) for any future change of model or data.